A new pairing for a core on eight points

Human Compression project: a component of the L044 recovery

Let the natural A_{2m} have a string C-marking of rank m, where $m \geq 8$ is even. Let $H = G_i$ be a transitive maximal omission preserving two blocks of size m, called its rows. Every distinguished maximal omission $H_j = G_{i,j}$ is assumed intransitive. Choose a mark l interchanging the rows. Suppose its two noncommuting neighbours in H are r, s, in that order around l, and let C be the other $m-4$ marks. Thus r, s commute, and every mark of C commutes with l and with at least one of r, s.

We acquire the proved contracted-forest and local-support lemmas. They supply one eligible selected edge for every mark other than l, and the full graph J of the contracted r, s edges has rank three. In this paper we impose the additional hypothesis that J is connected on exactly four l-pairs, denoted by Y. All other l-pairs are isolated in J. The reduction to this case is a separate obligation.

We assume the alternating rank bound $\lfloor (q-1)/2 \rfloor$ in every degree $12 \leq q < 2m$, and retain the exact acquired smaller-degree maxima. These are the induction hypotheses of the completed pair-block theorem. The complete transitive rank theorem and extremal symmetric independent-generator theorem are retained as well.

Theorem 1. *Under these hypotheses, if at least one of r, s interchanges the two rows, the configuration is impossible.*

The selected bridges and the two actual forms

Write the points as x_0, x_1, with l interchanging the two points in each pair. A mark in C has the form $\mathrm{diag}(\pi) l^{\chi}$, where π is an involution of the pair indices and $\chi \in \{0, 1\}$. This is a genuine action for C marks; no such action is asserted for r, s.

Lemma 2. *Each selected C edge has an endpoint outside Y. Every other contracted edge of that same mark lies in Y, and there is at most one such additional edge.*

Proof. By the acquired bridge lemma, J together with the $m-4$ selected C edges is connected, and every selected edge is a bridge. No such edge can have both endpoints in the connected graph on Y. Collapsing that graph to one vertex leaves a tree.

Take an additional contracted edge of a mark $g \in C$, and join its endpoints by a simple path in the connected graph. The path cannot contain a selected edge of another C label: the new edge would provide a bypass, which lifts using l to a walk in that label's actual omission, contradicting eligibility. If the selected g edge occurs, it cannot be the first or last edge of this path, because the distinct transpositions of the projected involution g are disjoint. Both remaining path parts then use only J, so both ends of that selected edge lie in Y, already excluded. Hence the path uses only J, and the additional edge lies in Y. The acquired local-support lemma allows at most two simple edges or a triangle for one label. A projected involution is a matching, so the triangle is impossible. □

Reverse the string if necessary so that r interchanges rows. For a permutation p of the pair indices, define

$$\mathrm{Swap}(p) : x_0 \longmapsto p(x)_1, \qquad x_1 \longmapsto p^{-1}(x)_0.$$

The acquired support lemma gives $r = \mathrm{Swap}(p)$ for a three-cycle p, with p fixing all other indices. Name the four core pairs a, b, c, d so that $p = (a\ b\ c)$.

Lemma 3. *After rotating a, b, c, precisely the following two forms remain:*

$$\begin{aligned} &\mathrm{M}: \quad s = (b_0\ d_0)(c_1\ d_1), \\ &\mathrm{S}: \quad s = \mathrm{Swap}((b\ d\ c)). \end{aligned} \tag{1}$$

In both forms, the involution

$$z = \mathrm{diag}((b\ c))l \tag{2}$$

centralizes r, l, s, interchanges the rows, and equals l outside Y.

Proof. If s preserves the rows, it is a double transposition by the acquired local-support lemma. Commutation with the row-swapping r forces one transposition in each row. If the first is t, the second is t^p, using right permutation multiplication. Since its edges and the triangle of p have union Y, the transposition t joins d to one of a, b, c. Rotate the latter indices to obtain $t = (b\ d)$ and $t^p = (c\ d)$.

If s interchanges the rows, the same support lemma gives $s = \mathrm{Swap}(q)$ for a three-cycle q. Its support meets that of p in two points, since their union has four points. Rotate a, b, c so that the support of q is $\{b, c, d\}$. Commutation is exactly $pq^{-1} = qp^{-1}$. Substituting the two orientations of this three-cycle leaves $q = (b\ d\ c)$; the other orientation does not satisfy the equation. Finally, direct substitution in these two forms proves every assertion about (2). □

Marks that might break the new pairing

Let $g = \mathrm{diag}(\pi)l^\chi$ be a mark in C. By Lemma 2, write $\pi = e\tau$, where e is its selected transposition and τ is either the identity or a single transposition in Y, disjoint from e.

Suppose first that g commutes with r. The six points over a, b, c form the unique six-point orbit of $\langle l, r\rangle$; all other orbits are pairs. Thus e is either $(d\ x)$ or $(x\ y)$, with x, y outside Y. On a, b, c, the permutation π centralizes p when $\chi = 0$ and inverts p when $\chi = 1$. Its involutory possibilities are therefore

$$(\tau, \chi) = (1, 0), \quad ((a\ b), 1), \quad ((a\ c), 1), \quad ((b\ c), 1). \tag{3}$$

Only the middle two possibilities fail to commute with z.

If g instead commutes with s in form M, its selected edge is $(a\ x)$ or is entirely outside Y. Commutation with the two row transpositions forces $(\tau, \chi) = (1, 0)$ or $((b\ c), 1)$: with no row interchange it preserves both transpositions, while with row interchange it exchanges them. Both possibilities commute with z.

In form S, commuting with s gives the analogue of (3) on b, c, d. The two failures are $((b\ d), 1)$ and $((c\ d), 1)$. Reverse r, s and rename

$$(a', b', c', d') = (d, c, b, a).$$

This preserves the form S and changes these failures into the middle cases of (3). Moreover conjugation by z fixes r, l, s and exchanges $(a\ b)$ with $(a\ c)$, leaving either possible selected edge unchanged. It therefore suffices in both forms to exclude

$$g = \mathrm{diag}(e(a\ b))l, \qquad e = (d\ x) \text{ or } e = (x\ y). \tag{4}$$

Every mark in C commutes with at least one of r, s, so this list covers all possible failures of centrality with z.

Lemma 4. *In form* M*, a mark (4) violates the actual intersection property.*

Proof. Put $s' = lsl$ and $t = lg$. The following uniform permutation identities use right multiplication:

$$w = \begin{cases} (ts')^2(st)^2s(s't)^2ss's, & e = (d\ x), \\ \big(l(sg)^2s\big)^2, & e = (x\ y). \end{cases} \tag{5}$$

In each case,

$$w = (a_0\ b_0\ c_0)(a_1\ c_1\ b_1) = rl. \tag{6}$$

For completeness, substituting $s = (b_0\ d_0)(c_1\ d_1)$, $s' = (b_1\ d_1)(c_0\ d_0)$ and the appropriate t into (5) gives the point images

v	a_0	b_0	c_0	a_1	b_1	c_1
v^w	b_0	c_0	a_0	c_1	a_1	b_1.

The points over d, x, y are fixed whenever present, and the global row interchanges cancel on every other pair. This verifies (6) on the entire domain. Hence $r = wl$ lies in $\langle l, s, g\rangle$, contradicting IP for the singleton mark r and the disjoint marked set $\{l, s, g\}$. No finite enumeration is needed. □

Lemma 5. *In form* S*, a mark (4) makes H_l transitive.*

Proof. The marks r, s, g connect all eight core points by the walk

$$a_0 \overset{r}{-} b_1 \overset{s}{-} c_0 \overset{r}{-} a_1 \overset{g}{-} b_0 \overset{r}{-} c_1 \overset{s}{-} d_0 \overset{r}{-} d_1.$$

All these are edges of the actual omission $H_l = \langle r, s, C\rangle$. On every outside pair, r equals l, so that pair also lies inside an H_l-orbit. Consequently all H_l-orbits are unions of whole l-pairs, and the eight core points lie in one orbit. The connected graph consisting of J and the selected C bridges now joins all these orbits. Every edge of J lies in the core orbit, and each selected edge is an actual edge of a mark of H_l. Thus H_l is transitive, contrary to hypothesis. □

Maximality of the new pair system

The preceding lemmas show that every mark of H centralizes z. Its orbits are pairs meeting the original rows once. In coordinates matching those rows by z, the group H therefore lies in $\mathrm{diag}(S_m) \times \langle z\rangle$. Let P be its transitive image on the new pairs.

If all $m-1$ projected distinguished marks are independent, the acquired extremal symmetric independent-generator theorem gives $P = S_m$. Otherwise delete a redundant projected mark j. The actual subgroup H_j still projects onto P, but is intransitive on the $2m$ points. Its two point orbits each meet every new pair once. Hence the projection is faithful and gives a genuine transitive string C-marking of P of rank $m-2$. For $m \geq 10$, the complete acquired transitive rank theorem forces $P = S_m$ or A_m, since $m-2 > m/2+1$. The complete degree-eight boundary excludes rank six for every proper transitive group other than A_8. Thus P is primitive in all cases, proving that the new pair system is maximal.

The completed pair-block theorem for the parent A_{2m} now excludes the configuration under its explicit smaller-degree induction. This proves Theorem 1. The argument nowhere requires z to belong to H; centrality as a permutation and the actual omission orbits suffice.

Attribution and scope. The historical target is part of Proposition 3.5 of Cameron–Fernandes–Leemans–Mixer. The acquired forest and bridge arguments were written by the root investigator, who also suggested seeking a new central pairing here. This continuation supplies the two actual forms, the uniform IP identities, the eight-point walk and the maximality argument. The connected four-pair hypothesis remains explicit; its reduction and the case where both neighbours preserve rows are separate components. No whole-source recovery is claimed by this paper.